

Skills

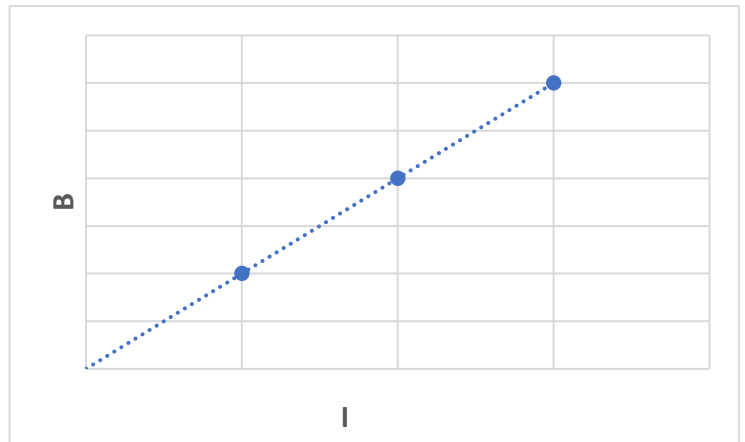
1- Slope

Suppose we have a relation like

$$\text{this } B = \frac{\mu I}{2\pi d}$$

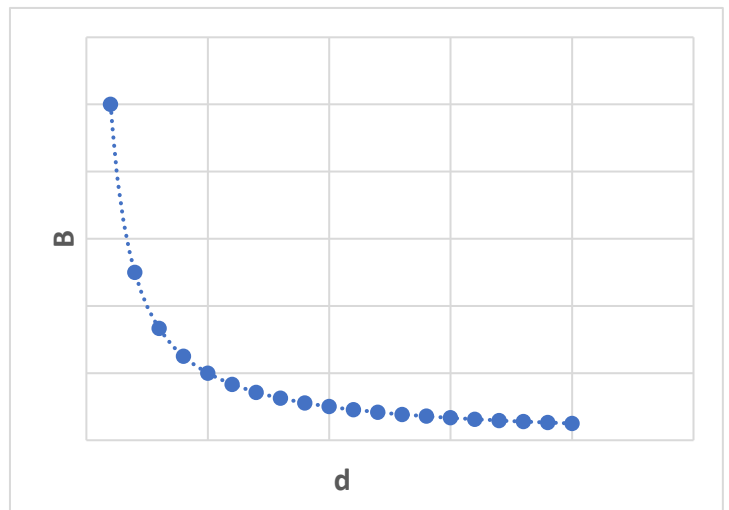
- To Plot a relation between B & I (direct)

$$\text{Slope} = \frac{B}{I} = \frac{\mu}{2\pi d}$$



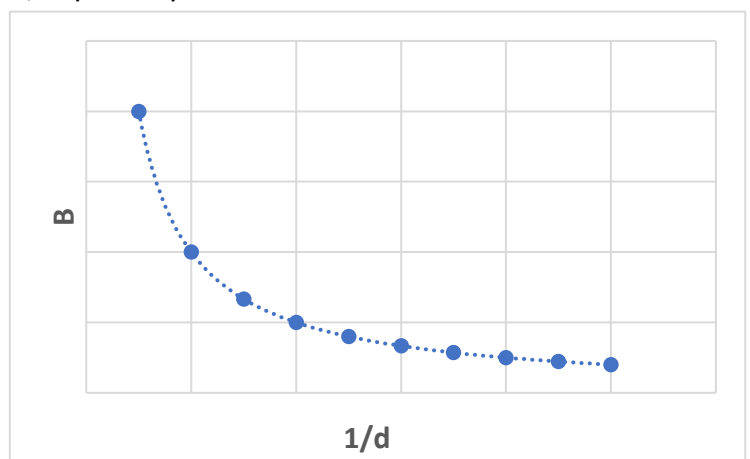
- To plot a relation between B and d (inverse)

No Slope



- To plot a relation between B and 1/d (direct)

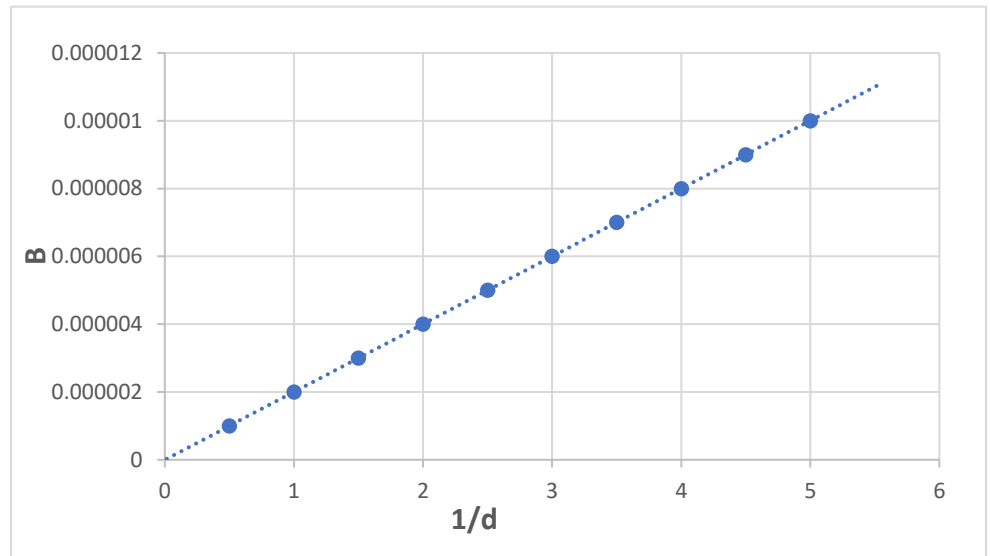
$$\text{Slope} = \frac{B}{\frac{1}{d}} = Bd = \frac{\mu I}{2\pi}$$



(Ex) from the following plot and given the relation $B = \frac{\mu I}{2\pi d}$

If $\mu = 4\pi * 10^{-7}$

Calculate I?



Solution:

$$\text{Slope} = \frac{B}{\frac{1}{d}} = Bd = \frac{\mu I}{2\pi}$$

$$\text{Slope} = \frac{\mu I}{2\pi}$$

$$\frac{0.0001}{5} = \frac{4\pi * 10^{-7} * I}{2\pi}$$

You can solve this equation using the calculator or by cross multiplication.

$$I = \frac{2\pi * 0.0001}{5 * 4\pi * 10^{-7}} = \dots\dots\dots$$

2- Ratio

Suppose we have a relation like this $B = \frac{\mu I}{2\pi d}$, and we want to make a ratio

$$\frac{B_1}{B_2} = \frac{\mu_1 I_1 d_2}{\mu_2 I_2 d_1}$$

المتغيرات الطردية تترتب نفس الترتيب والعكسية تترتب بالعكس
الثوابت مش بحطها في النسبة

(EX2) given this equation $B = \frac{\mu NI}{2r}$ if $\mu_1 = \mu_2$, $I_1 = 3I_2$, $N_1 = 2N_2$, $r_1 = \frac{1}{5} r_2$

find $\frac{B_1}{B_2}$?

- First make the ratio

$$\frac{B_1}{B_2} = \frac{\mu_1 I_1 N_1 r_2}{\mu_2 I_2 N_2 r_1}$$

- Then substitute with the givens.

$$\frac{B_1}{B_2} = \frac{\mu_1 3I_2 2 N_2 r_2}{\mu_2 I_2 N_2 \frac{1}{5} r_2}$$

$$\frac{B_1}{B_2} = \frac{\cancel{\mu_1} 3\cancel{I_2} 2\cancel{N_2} \cancel{r_2}}{\cancel{\mu_2} \cancel{I_2} \cancel{N_2} \frac{1}{5}\cancel{r_2}}$$

$$\frac{B_1}{B_2} = \frac{1*3*2}{1*1*\frac{1}{5}} = \frac{11}{1}$$

- Now suppose that $B_1 = 22$, calculate B_2 ?

$$\frac{22}{B_2} = \frac{11}{1}$$

$$B_2 = \frac{22}{11} = 2$$

3- factors affecting.

Suppose we have a relation like this $B = \frac{\mu I}{2\pi d}$

What are the factors affecting B?

1. μ (*direct*)
2. I (*direct*)
3. d (*inverse*)

Note: 2 and π are constants doesn't affect B

4- Increase or decrease?!

Suppose we have a relation like this $B = \frac{\mu I}{2\pi d}$

To increase B, we must

1. Increase μ (*direct*)
2. Increase I (*direct*)
3. Decrease d (*inverse*)

(EX3) if I increased to double what happen to B?

$$B \propto I$$

If $I \uparrow 2$ so $B \uparrow 2$

(EX4) if d increased to double what happen to B?

$$B \propto \frac{1}{d}$$

If $d \uparrow 2$ so $B \downarrow \frac{1}{2}$ (decrease to half)

(EX5) if d increased to double and I increased three times what happen to B?

$$B = \frac{\mu I}{2\pi d} = \frac{1 \cdot 3}{1 \cdot 1 \cdot 2} = \frac{3}{2}$$

So, B increase to 3/2 of its original value.

لاحظ بشيل الحاجه اللي بتتغير واحط مكانها الرقم
اللي اتغيرتله واللي مش بيتغير بحط مكانه واحد
الثوابت بتتقال ويتحط مكانها واحد مش بتتساب
زي ما هي